Curated NeuroVault: A Curated Meta-analytic Neuroimaging Dataset from NeuroVault

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VANDERBILT School *of* Medicine

OVERVIEW & MOTIVATION

NeuroVault is an open repository for sharing statistical maps, parcellations, and atlases of the human brain. Investigators worldwide upload group-level images for interactive visualization and large-scale voxel-level meta-analysis.
Raw images are highly heterogeneous: metadata is manually entered and error-prone, images span incompatible coordinate spaces and resolutions, and statistical maps are mixed with individual contrasts. These issues make direct meta-analytic use unreliable without systematic curation.

Goal: Build a reproducible, automated monthly pipeline that curates NeuroVault into a clean, analysis-ready dataset — standardized to MNI 2mm, converted to RESI effect sizes, and free of outlier images. Dataset released publicly on Zenodo; utility demonstrated via unsupervised and supervised meta-analyses in our preprint.

FULL PIPELINE AT A GLANCE

Metadata Download	Metadata Exclusions	Image Download	Image Exclusions	Registration DOF=6	Coverage QC	RESI Convert	LMM Outliers
618,582	→ ~23K	NIfTI	Dups, range	MNI 2mm	GM/WM	Effect size	Clean

618K	~23K	~23K	21,311
Raw NeuroVault images	After metadata exclusions	After image-level exclusions	Final curated dataset

~10% of NeuroVault images are valid group-level statistical maps. Including only T/Z maps is the single largest reduction (~70%). Pipeline auto-runs on the 1st of each month via R Markdown; completed steps cached as monthly-versioned RDS files.

STEP 1 METADATA EXCLUSIONS

Study- and image-level metadata downloaded from the NeuroVault API (paginated collections + image endpoints). Six exclusionary criteria applied to 618K images:

ANALYSIS_LEVEL = GROUP
Retain only group-level maps; individual-level contrasts excluded. Group images are <10% of all entries.

NOT_MNI = FALSE
Exclude images not in MNI space. Largest spatial filter after map-type selection.

IS_THRESHOLDED = FALSE
Keep only unthresholded maps — thresholding introduces significant-result bias.

MAP_TYPE ∈ {T, Z}
Restrict to T and Z statistics. Removes F maps, beta maps, variance maps.

NUMBER_OF_SUBJECTS > 0
Exclude missing/zero sample size. Required for RESI conversion.

N ≤ 100,000
n=32,222,222 spot-checked (true n=32, corrected). Values >100K flagged as entry errors.

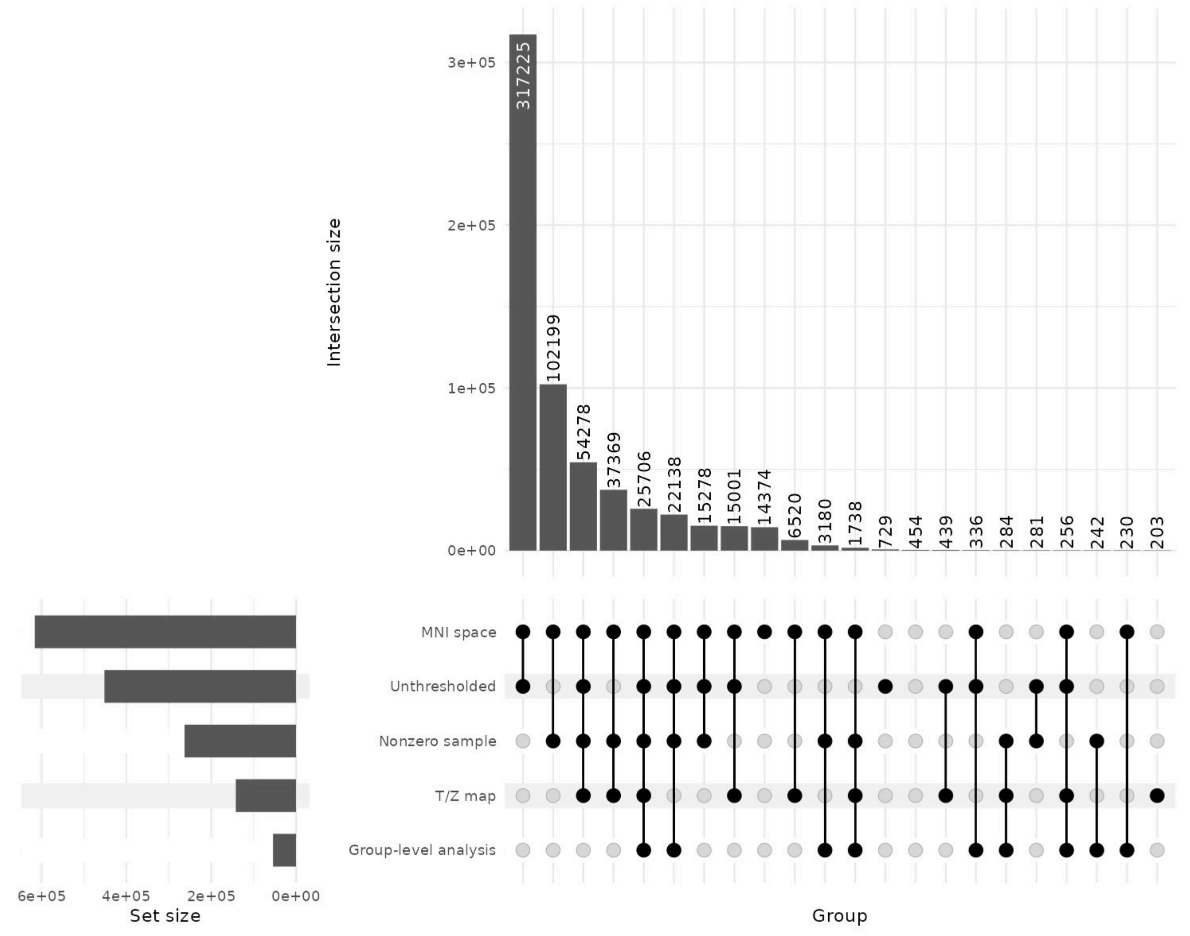


Fig 1. UpSet plot of exclusionary criteria (618,582 images). Largest intersection (317,225): thresholded only. Second (102,199): non-MNI only. Filled dots = criterion satisfied; connected = co-occurring failures.

STEP 2 IMAGE-DERIVED EXCLUSIONS

After NIfTI download, five properties computed per image via `3d -info`: `imgdim`, `voxdim`, `nvox`, `range`, `bounding_box`.

① **Duplicates:** Images sharing identical range AND filename basename flagged; first occurrence retained. Two distinct images sharing both attributes is exceedingly rare.

② **Proportionality:** Physical dims (mm) = imgdim × voxdim. Each axis checked against MNI template (182×218×182 mm) with per-axis ratio bounds [0.75, 1.25]. 23,191 of 23,594 pass.

③ **Range exclusion:** |max − min| < 0.01 ($\log_{10}$ < −2) indicates no intensity variation — not a valid statistical map.

Value processing: NaN → 0; voxels with |v| < ε=0.001 zeroed (prevents FLIRT failures). ε derived from 90th quantile of values outside an eroded non-brain mask.

STEP 3 REGISTRATION TO MNI 2MM

Despite self-reported MNI space, visual inspection revealed systematic misalignment. Three strategies evaluated on the 7 most common image dimension categories:

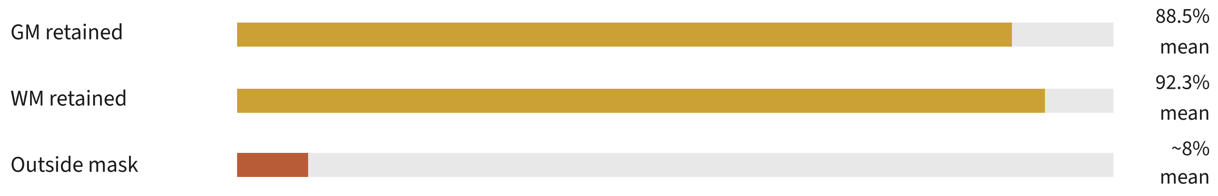
Reslicing only
Resample to MNI 2mm grid; no registration. Fails for misaligned images — doesn't correct coordinate offsets.

Rigid (DOF=6) ✓✓
Rotation + translation only. Preserves brain shape and scale. Best accuracy/flexibility balance.

Affine (DOF=12)
Adds scaling + shearing. Over-flexible — can distort effect size geometry without improving alignment.

FSL FLIRT with mutual information cost (`-cost mutualinfo -interp trilinear`). All output images: 2mm isotropic, 91×109×91 voxels.

Post-registration coverage QC: GM, WM, and outside-brain voxel overlap computed for each registered image. Thresholds: GM > 100K voxels, WM > 20K voxels, outside < 100K voxels. Low GM/WM = partial coverage or failed registration; high outside = absent skull-stripping.



STEP 4 EFFECT SIZE ESTIMATION (RESI)

Nonzero voxels in T and Z images converted to the **signed Robust Effect Size Index (RESI)** using each image's sample size *n*:

→ T maps: `RESI::t2S(t, rdf = n−1, n = n)`
→ Z maps: `RESI::z2S(z, n = n, unbiased = TRUE)`

RESI is sample-size-independent, enabling valid cross-study comparison regardless of heterogeneous N. Unlike Cohen's d, RESI is defined for all inferential tests and remains bounded in expectation. A brain-masked version (`*_masked_mni.nii.gz`) zeros non-brain voxels for downstream meta-analysis.

Output variable	Description
`es_mean`	Mean RESI over all nonzero voxels
`grey_mean / white_mean`	Mean RESI within GM/WM masks
`es_min/median/max`	Quantile summaries (0, .25, .5, .75, 1)
`voxel_scaled`	Nonzero voxel count ÷ 1,000

STEP 5 META-ANALYTIC OUTLIER DETECTION (LMM)

Mean effect size $\bar{y}_{ij}$ for image *j* in collection *i* is modelled with a heteroscedastic mixed-effects model; residual variance scales with $n_i^{-1} V_{ij}^{-2}$:

$$\bar{y}_{ij} = \mu + u_i + \varepsilon_{ij}, \quad u_i \sim N(0, \tau^2)$$
$$Var(\varepsilon_{ij}) = \sigma^2 \cdot |N_{ij}|^{2\delta_n} \cdot |V_{ij}|^{2\delta_v}$$

Outlier criterion: $|\bar{y}_{ij} - \hat{\mu}| > k \cdot \hat{\sigma}_{ij}$, k=3, where $\hat{\sigma}^2_{ij} = \hat{\tau}^2 + \hat{\sigma}^2 \cdot N_{ij}^{2\hat{\delta}_n} \cdot V_{ij}^{2\hat{\delta}_v}$. Model fit on **Winsorized data** (p1–p99) first to prevent extreme images from distorting their own exclusion boundary, then refit on cleaned data. $\hat{\delta}_n < 0$ and $\hat{\delta}_v < 0$ confirm larger/denser images have lower variance as theoretically expected.

Model	Dataset	Purpose
Original	Full raw data	Baseline fit
Winsorized	p1–p99 clipped	Outlier boundaries
Cleaned	Outliers removed	Final estimate

FINAL CURATED DATASET & OUTPUTS

Stage 1: Metadata exclusions (group-level, MNI, T/Z, unthresholded, n>0)	−595K
Stage 2: Image-derived exclusions (duplicates, proportionality, range <0.01)	−~400
Stage 3: Coverage exclusions (GM, WM, outside-brain thresholds)	−~1,400
Stage 4: LMM outlier exclusions ($|ES - \hat{\mu}| > 3 \cdot \hat{\sigma}$, Winsorized model)	−~500
✓ Final Curated Dataset — 21,311 images, ~4,500 collections	21,311

Monthly Zenodo release contents:

→ Tabular RDS (`curatedMetadataset_YYYY-MM.rds`): one row per image, all QC metrics and effect size summaries
→ Brain-registered RESI NIfTI images in MNI 2mm (`*_effect_size_masked_mni.nii.gz`)
→ Effect size summaries (mean + quintiles) within GM & WM per image

RESULTS LMM PARAMETER ESTIMATES

Three model variants confirm stability of the outlier detection procedure across curation stages. Reductions in $\hat{\tau}$ and $\hat{\sigma}$ from Original → Cleaned confirm curation successfully removed noisy images.

Parameter	Original	Winsorized	Cleaned
$\hat{\mu}$ (grand mean ES)	—	—	—
$\hat{\tau}$ (between-collection SD)	High	Med	Low ↓
$\hat{\sigma}$ (residual SD)	High	Med	Low ↓
$\hat{\delta}_n$ (sample size power)	< 0	< 0	< 0
$\hat{\delta}_v$ (voxel coverage power)	< 0	< 0	< 0

$\hat{\delta}_n < 0$ and $\hat{\delta}_v < 0$ across all models: larger studies and images with denser voxel coverage show smaller effect-size variance, consistent with theoretical derivation. Fill in exact values from your `intervals(m_power)` output.

FINAL DATASET SUMMARY (N=21,311)

Variable	Mean (SD)	Median [Min, Max]
`number_of_subjects`	81.8 (240)	52 [3, 30000]
`es_mean`	0.025 (0.845)	0.004 [−52.7, 45.9]
`voxel_scaled`	240 (43.9)	228 [130, 344]
`gm_perc`	0.885 (0.099)	0.902 [0.56, 1.00]
`wm_perc`	0.923 (0.090)	0.944 [0.34, 1.00]
`outside_perc`	0.033 (0.038)	0.010 [0, 0.15]
`grey_mean`	0.028 (0.971)	0.006 [−64.9, 52.4]
`white_mean`	0.020 (0.763)	0.002 [−43.2, 38.2]

609 outliers (2.8% of 21,920) removed by LMM criterion. GM coverage 88.5% and WM coverage 92.3% confirm adequate brain signal across retained images.

KEY VARIABLES & REPRODUCIBILITY

OUTPUT VARIABLES

`es_mean` Mean RESI, all voxels
`grey/white_mean` Masked RESI
`gm/wm_perc` Coverage %
`voxel_scaled` Voxels ÷ 1K
`collection_id` LMM grouping

RUNTIME (24 CORES)

Metadata download: ~30 min
Image download: ~2 hr
Registration (FLIRT): ~4 hr
Effect sizes (RESI): ~6 hr
Outlier LMM: <5 min

DATASET ACCESS & CITATION

The curated dataset is publicly available on **Zenodo** and updated automatically on the 1st of each month. Each release includes tabular metadata (RDS) and brain-registered RESI effect size images (NIfTI, MNI 2mm).

DOI
https://doi.org/10.5281/zenodo.20293389

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MONTHLY RELEASE CONTENTS

→ `curatedMetadataset_YYYY-MM.rds` — tabular metadata, one row per image with all QC metrics and effect size summaries
→ `neurovault_effectsizes_masked_YYYY-MM.tar` — brain-registered RESI NIfTI images in MNI 2mm space
→ Effect size summaries (mean + quintiles) within gray & white matter per image

CONTACT & CODE

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Pipeline code & R Markdown source:
`github.com/statimagcoll/statimagcoll.github.io`

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